PHILOSOPHICAL TRANSACTIONS A

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1. Insert A head here

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Subsubsection text here.

2. Equations

Sample equations.

$$\begin{aligned}\frac{\partial u(t,x)}{\partial t} &= Au(t,x)\left(1-\frac{u(t,x)}{K}\right) - B\frac{u(t-\tau,x)w(t,x)}{1+Eu(t-\tau,x)},\\ \frac{\partial w(t,x)}{\partial t} &= \delta\frac{\partial^2 w(t,x)}{\partial x^2} - Cw(t,x) + D\frac{u(t-\tau,x)w(t,x)}{1+Eu(t-\tau,x)},\end{aligned} \tag{2.1}$$

$$\begin{aligned}\frac{dU}{dt} &= \alpha U(t)(\gamma - U(t)) - \frac{U(t-\tau)W(t)}{1+U(t-\tau)},\\ \frac{dW}{dt} &= -W(t) + \beta\frac{U(t-\tau)W(t)}{1+U(t-\tau)}.\end{aligned} \tag{2.2}$$

$$\begin{aligned}\frac{\partial(F_1,F_2)}{\partial(c,\omega)}_{(c_0,\omega_0)} &= \left|\begin{array}{cc}\frac{\partial F_1}{\partial c} & \frac{\partial F_1}{\partial \omega}\\ \frac{\partial F_2}{\partial c} & \frac{\partial F_2}{\partial \omega}\end{array}\right|_{(c_0,\omega_0)}\\ &= -4c_0q\omega_0 - 4c_0\omega_0p^2 = -4c_0\omega_0(q+p^2) > 0.\end{aligned} \tag{2.3}$$

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3. Enunciations

Theorem 3.1. *Assume that* $\alpha > 0, \gamma > 1, \beta > \frac{\gamma+1}{\gamma-1}$. *Then there exists a small* $\tau_1 > 0$, *such that for* $\tau \in [0, \tau_1)$, *if* c *crosses* $c(\tau)$ *from the direction of to a small amplitude periodic traveling wave solution of (2.1), and the period of* $(\check{u}^p(s), \check{w}^p(s))$ *is*

$$\check{T}(c) = c \cdot \left[\frac{2\pi}{\omega(\tau)} + O(c - c(\tau)) \right].$$

Condition 3.1. *From (0.8) and (2.10), it holds* $\frac{d\omega}{d\tau} < 0$, $\frac{dc}{d\tau} < 0$ *for* $\tau \in [0, \tau_1)$. *This fact yields that the system (2.1) with delay* $\tau > 0$ *has the periodic traveling waves for smaller wave speed* c *than that the system (2.1) with* $\tau = 0$ *does. That is, the delay perturbation stimulates an early occurrence of the traveling waves.*

4. Figures & Tables

The output for figure is:

Figure 1. Insert figure caption here

The output for table is:

Table 1. An Example of a Table

date	Dutch policy	date	European policy
1988	Memorandum Prevention	1985	European Directive (85/339)
1991–1997	**Packaging Covenant I**		
1994	Law Environmental Management	1994	European Directive (94/62)
1997	Agreement Packaging and Packaging Waste		

5. Conclusion

The conclusion text goes here.

Acknowledgements. Insert acknowledgment text here.

References

1. Allwood JM, Cullen JM. 2011 *Sustainable materials: with both eyes open*. Cambridge, UK: UIT Cambridge. See http://www.withbotheyesopen.com.
2. MacKay DJC. 2008 *Sustainable energy: without the hot air*. Cambridge, UK: UIT Cambridge. See http://www.withouthotair.com.
3. Gallman PG. 2011 *Green alternatives and national energy strategy: the facts behind the headlines*. Baltimore, MD: Johns Hopkins University Press.
4. MacKay DJC. 2013. Solar energy in the context of energy use, energy transportation, and energy storage. *Proc. R. Soc. A* **371**.